

Manufacturer		Customer	Business Transaction			
Steel Heat Treatment GmbH 123 Industrial Park 40210 Düsseldorf DE		Automotive Components Inc 456 Manufacturing Drive Detroit 48220 US	Order	ORD-2024-HT-047		
			Delivery	DEL-2024-047 · 4750 kg		
			Product			
			Product Name	Carbon Steel Round Bar 42CrMo4	Name (EN 10027-1)	42CrMo4
			Batch ID	BATCH-2024-047	Number (EN 10027-2)	1.7225
			Delivery Condition	+QT - Quenched and Tempered	Name (AISI/SAE)	4140
					Shape	RoundBar - D 80 mm - Length 6000 mm
			Heat Treatment			
			Process: Quenching and Tempering - Lot: HT-2024-11-15-B47 - Furnace: FURNACE-03 - Date: 2024-11-15			
Stage	Temperature	Duration	Cooling	Atmosphere		
Austenitizing	850 C	45 min		Air		
Quenching	60 C	10 min	Oil			
Tempering	550 C	120 min	Air			
Mechanical Properties						
Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength	Rm	985 MPa	900 MPa	1100 MPa	EN ISO 6892-1	✓
Yield Strength	Re	835 MPa	750 MPa		EN ISO 6892-1	✓
Elongation	A	14.5 %	10.0 %		EN ISO 6892-1	✓
Rockwell Hardness	HRC	33 HRC	28 HRC	35 HRC	ASTM E18	✓
Validation						
We hereby certify that the material described above has been heat treated and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.				Validated by Dr. Klaus Schmidt , Quality Manager,		
				Quality Assurance - 2024-11-15		